Foundations of Deep Learning

Introduction to Deep Learning

Introduction to Deep Learning

Frank Hutter Abhinav Valada

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Lecture Overview

1. Motivation for Deep Learning
2. Representation Learning and Deep Learning
3. Standard ML Concepts We Build On
4. Basics of Neural Networks
5. Flexibility of Deep Learning
6. History of Deep Learning
7. Summary, Further Reading, References

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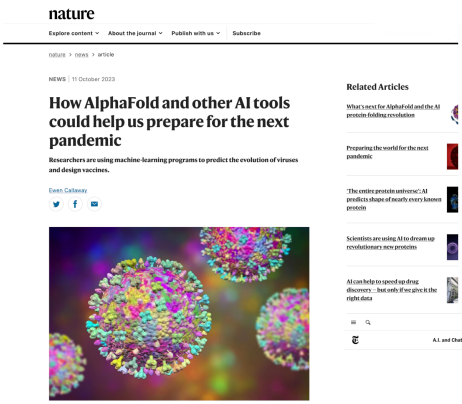
Motivation for Deep Learning

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Deep Learning is *The* Key Technology of Our Time


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NEWS | 11 October 2023

How AlphaFold and other AI tools could help us prepare for the next pandemic

Researchers are using machine-learning programs to predict the evolution of viruses and design vaccines.

Ewen Callaway

The rare Langya virus (computer-generated image) is related to some deadly pathogens. Credit: Kateryna Kon/Science Photo Library

In the third year of the COVID-19 pandemic, researchers reported another worrying virus. Identified in 35 people in eastern China since 2018, Langya henipavirus causes breathing problems, fever and other troubling symptoms.

Langya hasn't been linked to any deaths yet. But it is related to some deadly viruses, so researchers were keen to develop vaccines. There was just one problem: a viral protein that could form the basis of a jab seemed impossible to make in the lab.

"If we can't even study the protein, how are we going to understand how it works, and how are we going to make a vaccine?" says David Veesler, a structural virologist at the University of Washington (UW) in Seattle.

RELATED

Now, thanks to artificial intelligence (AI), biologists have cracked Langya's nut. In work posted on the bioRxiv preprint server in August, Veesler's team describes using the prediction tool AlphaFold to map the structure of a protein with which Langya invades cells[1]. Another AI tool identified mutations that

Related Articles

What's next for AlphaFold and the AI protein-folding revolution

Preparing the world for the next pandemic

'The entire protein universe': AI predicts shape of nearly every known protein

Scientists are using AI to dream up revolutionary new proteins

AI can help to speed up drug discovery — but only if we give it the right data


KI-REGULIERUNG

Scholz und Macron träumen von EU-ChatGPT

VON HENDRIK WIEDUWILT · AKTUALISIERT AM 03.10.2023 · 19:57

Von wegen Brückentag: Für Lobbyisten ist dies die vielleicht härteste Woche des Jahres. Die EU will nämlich unbedingt ein Stück vom KI-Kuchen, und in der kommenden Woche droht der Anschnitt in Hamburg. Viele Hände führen nun die


The New York Times

A.I. and Chatbots › Meet the New ChatGPT · Meta's A.I. Characters · ChatGPT's Image Generator · Google's Bard Extensions · How Schools Can Survive A.I.


THE SHIFT

A Coming-Out Party for Generative A.I., Silicon Valley's New Craze

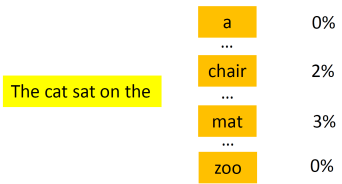
A celebration for Stability AI, the start-up behind the controversial Stable Diffusion image generator, represents the arrival of a new A.I. boom.

[Source: Articles on Nature, FAZ, and The New York Times]

Deep Learning Revolutionized Natural Language Processing

ChatGPT showed incredible results that surprised everyone

- Simply trained to predict the next word (plus fine-tuning to align with human values):



- To predict the next word as well as possible, it helps to "understand" the world
- GPT can understand any language, including code, use tools (plug-ins), etc

- General purpose technology with universal interface; extremely pervasive and disruptive

Deep Learning Revolutionized Generative AI

- Excellent results in generating many different types of output
  - images & videos
  - poetry
  - music
  - text to speech
  - code
  - proteins
  - ...
- Once more, pervasive and disruptive



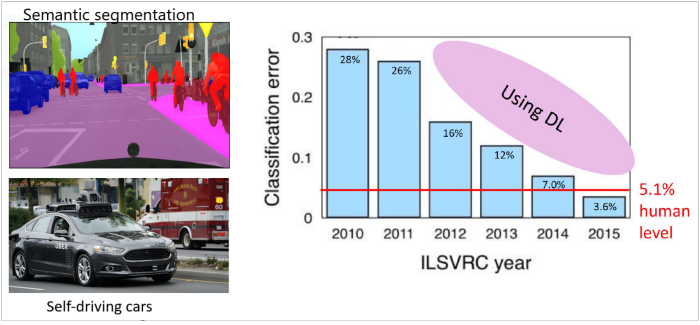
Generated by Midjourney

("portrait and paint drops")

[Image Sources: user Oronbb under CC license via Wikimedia Commons]

Deep Learning Revolutionized Computer Vision

- Excellent empirical results



ILSVRC: ImageNet Large-Scale Visual Recognition Challenge

[Image Sources: Valada et al, 2018, Dllu via Wikimedia Commons]

Deep Learning Revolutionized Speech Recognition

- Excellent empirical results

Speech recognition



Skype

Auto-Translator

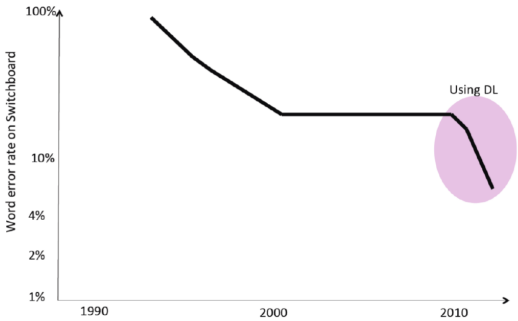


Image credit: Yoshua Bengio (data from Microsoft speech group)

[Image Sources: Dllu via Wikimedia Commons, pexels.com]

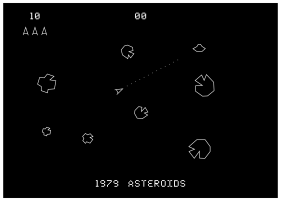
Deep Learning Goes Great with Reinforcement Learning

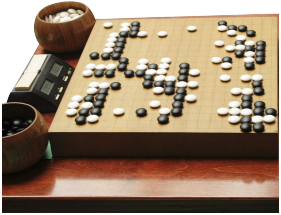
- Excellent empirical results obtained by deep reinforcement learning

  - Superhuman performance in playing Atari games...

  - and Superhuman performance in playing Go...

  - using one general algorithm
    [Schrittwieser et al., 2020]





[Image Sources: openclipart.org, HermanHiddema at the English Wikipedia via Wikimedia Commons]

(Deep) Learning as A Different Way of Programming

- We don't understand how the human brain solves certain problems such as:
  - Face recognition
  - Playing Atari games
  - Speech recognition
  - Picking the next move in the game of Go
- We can nevertheless learn these tasks from data/experience
- If the task changes, we simply re-train

Deep Learning Allows Many Branches of AI to Converge

- Deep learning is now the principle approach in many different branches of AI:
  - Computer vision
  - Speech recognition
  - Natural language processing
  - (Robotics)

- The same general techniques apply in all of these fields
  - Amazing potential for cross-fertilization
  - Fields that drifted apart for decades have largely converged again
  - E.g., in Freiburg:
    - close collaboration & joint reading group between machine learning, representation learning, computer vision, robotics, neurorobotics, and robot learning

Further Reasons for the Popularity of Deep Learning

- Very quick to get good results for some problems:
  - Deep learning can handle raw data (images, speech, text, etc).
  - Well-engineered libraries handle the complex underpinnings (TensorFlow, PyTorch, ...).
  - Very little machine learning knowledge is required to get started.
- Misconception: "it works like the brain"
- Neural networks are very flexible models – details discussed later

Questions to Answer for Yourself / Discuss with Friends

- Repetition: Why is deep learning so popular?
- Relation to your interests:
  Which problem might you want to apply deep learning to?
  - Think well; there will be a couple of follow-up questions on this

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Representation Learning and Deep Learning

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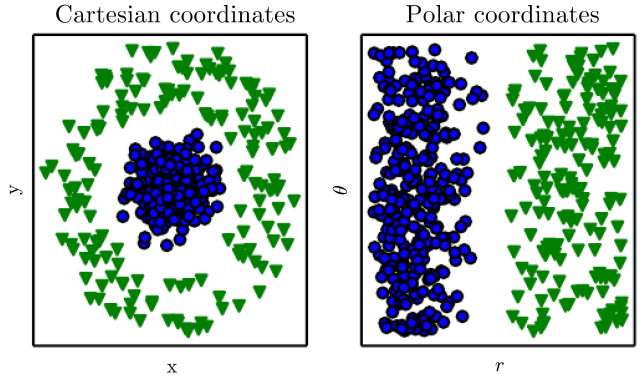
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Definition: Representation Learning

Representation Learning

"a set of methods that allows a machine to be fed with raw data and to automatically discover the representations needed for detection or classification" [LeCun et al., 2015]

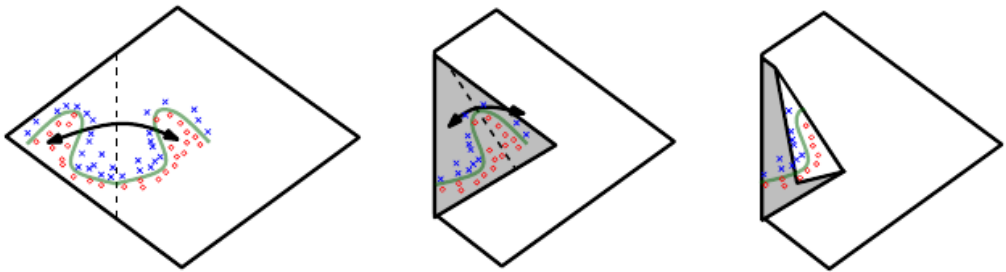


[Goodfellow et al., 2016, p. 4, fig 1.1]

Definition: Deep Learning

Deep Learning

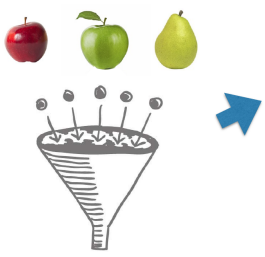
"representation learning methods with multiple levels of representation, obtained by composing simple but nonlinear modules that each transform the representation at one level into a [...] higher, slightly more abstract (one)" [LeCun et al., 2015]



[Goodfellow et al., 2016, p. 196, fig 6.5]

Standard Machine Learning Pipeline

- Standard machine learning algorithms are based on high-level attributes or features of the data
- E.g., the binary attributes we would use for decision trees
- This requires (often substantial) feature engineering

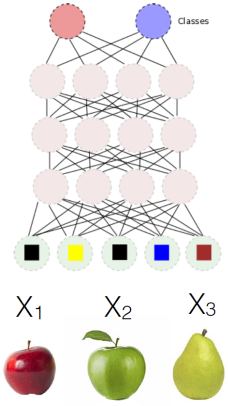


Features		Label
red	3.5 cm	apple
green	4 cm	apple
yellow	5cm	pear

[Image credit: Tobias Springenberg]

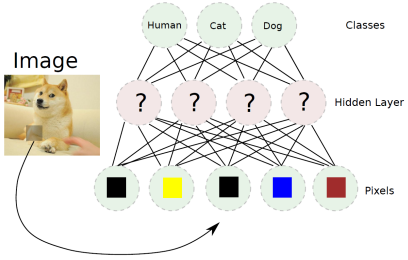
Representation Learning Pipeline

- Jointly learn features and classifier, directly from raw data
- This is also referrred to as end-to-end learning



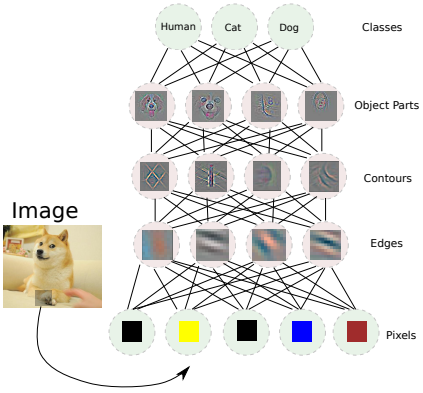
[Image credit: Tobias Springenberg]

Shallow vs. Deep Learning



[Image credit: Tobias Springenberg]

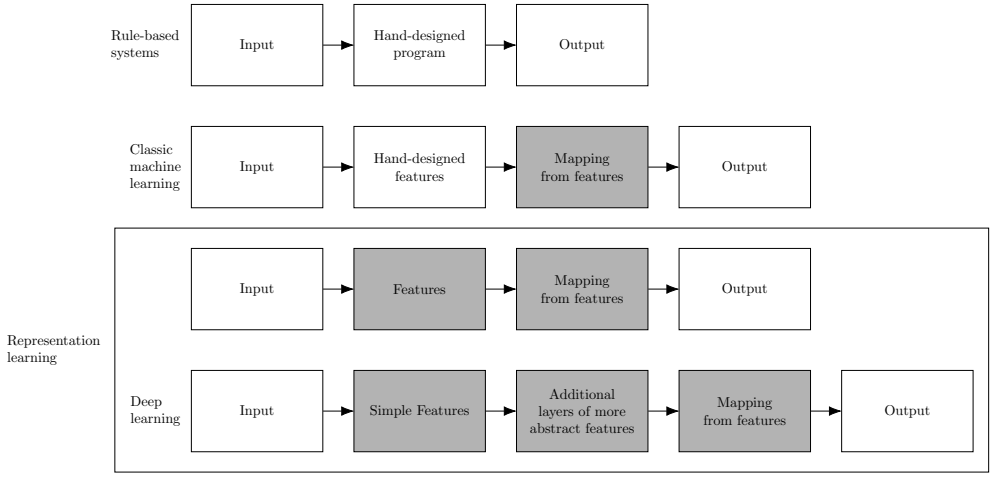
Shallow vs. Deep Learning



- Deep Learning: learning a hierarchy of representations that build on each other, from simple to complex
- Features are learned in an end-to-end fashion, from raw data

[Image credit: Tobias Springenberg. Visualizations of network activations taken from Zeiler et al., 2014]

Relation to More Traditional Learning Approaches



[Image Source: Goodfellow et al., 2016, adapted from p. 10, fig 1.5]

Questions to Answer for Yourself / Discuss with Friends

- Application to your interests:
  What would be a good and a bad representation for your problem of interest?

- Transfer: Are deep networks always better than shallow ones?

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Standard ML Concepts We Build On

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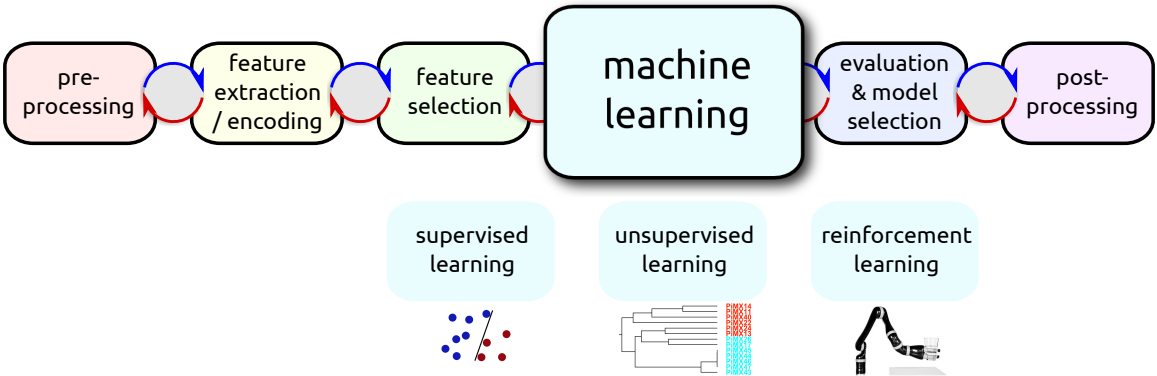
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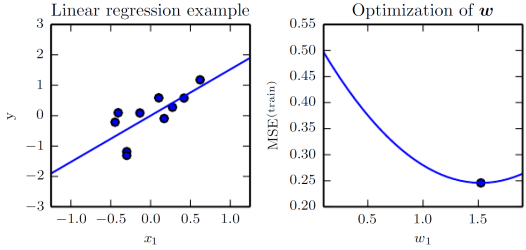
Prerequisites

- Good knowledge of linear algebra
- Good knowledge of machine learning, as, e.g., gained in our machine learning course

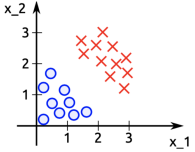
The Machine Learning Design Cycle



Linear Regression, Logistic Regression



[Goodfellow et al., 2016, p. 107, fig 5.1]



$$h_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}_0 + \mathbf{w}_1\mathbf{x}_1 + \mathbf{w}_2\mathbf{x}_2)$$

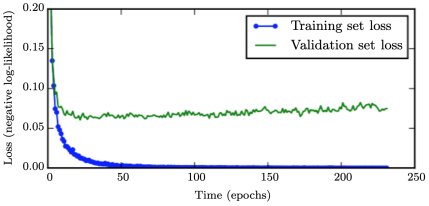
$$\mathbf{w} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

$$J(h_{\mathbf{w}}(\mathbf{x}), y) = \begin{cases} -\log(h_{\mathbf{w}}(\mathbf{x})) & \text{for} \quad y = 1 \\ -\log(1 - h_{\mathbf{w}}(\mathbf{x})) & \text{for} \quad y = 0 \end{cases}$$

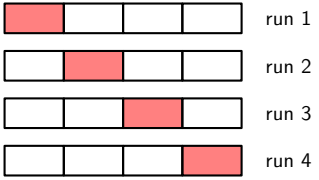
Minimizing loss functions with gradient descent

Overfitting, Training/Validation/Test, Cross-Validation

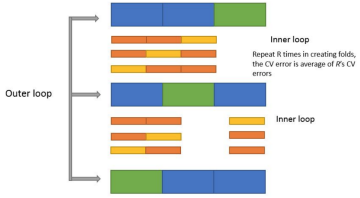
Over-fitting



Cross-validation



Inner cross-validation for tuning



Final evaluation only on a "fresh" test set

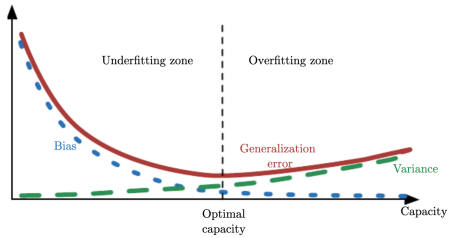
- Do not tune on the test set! This will seriously affect your grade.

[Image Sources: Goodfellow et al., 2016, p. 243, fig 7.3, Zhong et al., 2020]

Statistical Learning Theory

- Bias-Variance analysis to decompose generalization error (shown here for MSE)

$$\text{expected loss} = (\text{bias})^2 + \text{variance} + \text{noise}$$



- Vapnik-Chervonenkis dimension
  - Bound on the generalization error, typically far too large for deep learning, thus not practically useful

[Image Source: Goodfellow et al., 2016, p. 128, fig 5.6]

Linear Algebra

- Deep learning relies on fast matrix operations
  - In fact, tensor operations (tensors are like matrices, but have an arbitrary number of dimensions)
  - Almost all the math is written in matrix/vector form
  - E.g., backpropagation in matrix form is heavy on matrix algebra
- Some example concepts we'll require
  - Eigenvectors, eigenvalues, eigendecomposition
  - Hessians and Jacobians
- If you've gotten rusty, please refresh your skills, e.g., using:
  - `https://www.coursera.org/learn/matrix-algebra-engineers`
  - `https://www.khanacademy.org`

Python

- All our coding will be in Python
- We focus on foundations: deep understanding by implementing elements of deep learning from scratch
  - You need to know the basics: Python, Numpy, Matplotlib
    We will also use GitHub classroom
  - If you do not know these yet, please catch up using one of the *many* tutorials online, e.g.:
    `https://colab.research.google.com/github/kuleshov/teaching-material/blob/master/tutorials/python/cs228-python-tutorial-3_7.ipynb`
  - The first assignment will level the playing field w.r.t. Python (and Git)

- After understanding the basics, we'll switch to PyTorch
  - You do *not* need to know PyTorch before

Questions to Answer for Yourself / Discuss with Friends

- Repetition: How did logistic regression work exactly?
  (We'll need this next week)

- Self-assessment:
  Of the topics in this video, do you need to brush up on anything?

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Basics of Neural Networks

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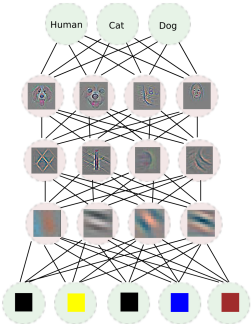
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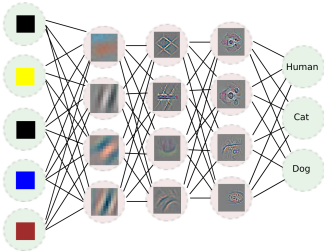
Canonical Ways of Depicting Neural Network Models

- There are different ways to draw a network
- The following three representations are most common:

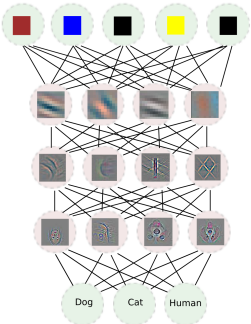
Bottom to Top:



Left to Right:

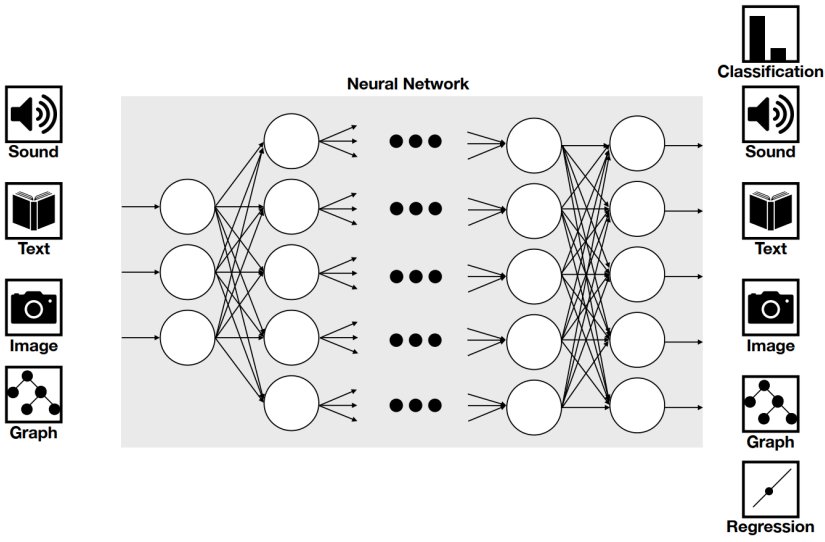


Top to Bottom:



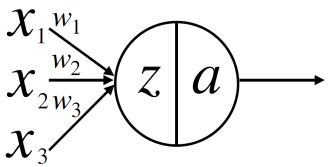
[Image credit: Tobias Springenberg]

Types of Layers in a Feedforward Neural Network



Computations in a Single Neuron

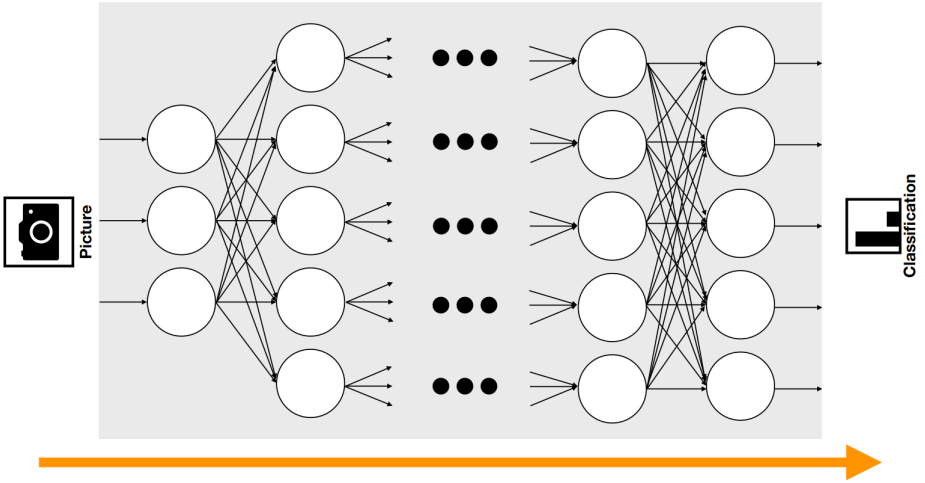
- A single neuron performs two simple steps of computation:



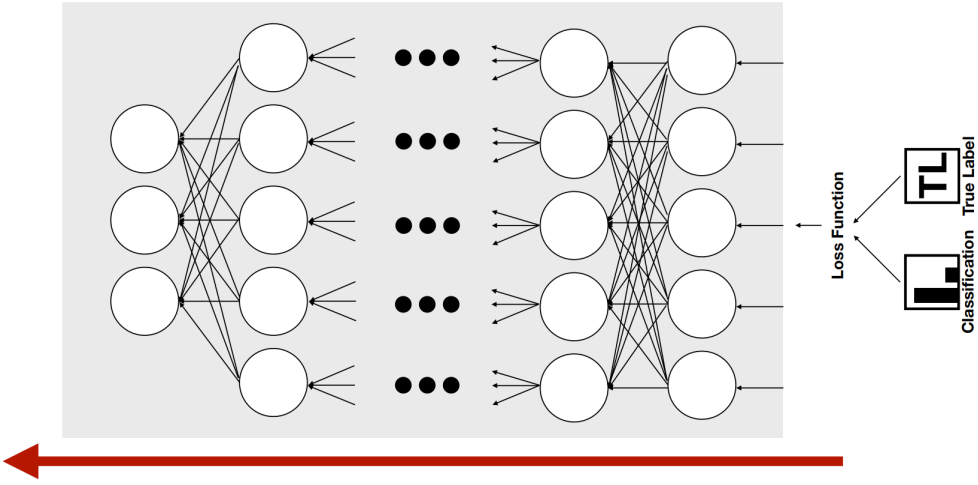
1. Compute a weighted sum of the inputs: $z = x_1 w_1 + x_2 w_2 + x_3 w_3$

2. Perform a nonlinear transformation: $a = h(z)$.

Information Flow in Fully-Connected Neural Nets – Forward Pass



Computing Gradients in Fully-Connected Neural Nets – Backward Pass



Questions to Answer for Yourself / Discuss with Friends

- Repetition: What computations are performed in a single neuron?
- Application of what you just learned:
  For a fully-connected feed-forward neural network with N layers of size K each, what is the computational complexity of the forward pass in big-O notation?
  (Hint: Visualize the MLP and/or the Matrix Multiplications)

If you don't know big-O notation, please look it up:
`https://en.wikipedia.org/wiki/Big_O_notation`

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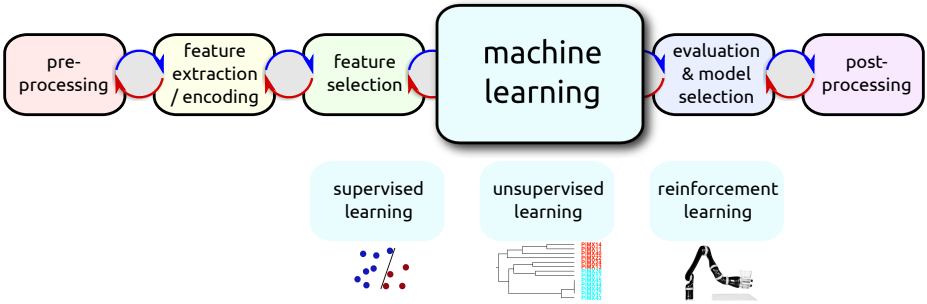
Flexibility of Deep Learning

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Deep Learning and the ML Design Cycle



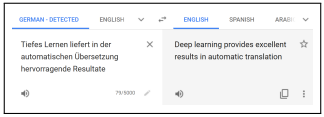
- Much reduced need for preprocessing
- No more need for manual feature engineering (extraction & selection)
  - useful features are learned automatically
- End-to-end training:
  - replacing most of the ML pipeline by a single approach
  - directly optimizing a single loss function

Flexibility of Deep Learning: Example Tasks

- Classification: input $\mathbb{R}^n \rightarrow$ categories (e.g., {cat, dog, human})
- Regression: input $\mathbb{R}^n \rightarrow \mathbb{R}$ (e.g., house prizes)
- Structured output prediction: structured output (vector, graph, etc)
  - Pixel-wise segmentation: image $\rightarrow$ image
  - Machine translation: sequence $\rightarrow$ sequence
  - Parsing: text $\rightarrow$ tree whose nodes are verbs, nouns, adverbs, etc
  - Image captioning: image $\rightarrow$ sentence
  - Denoising: corrupted example $\tilde{x} \in \mathbb{R}^n \rightarrow$ clean example $x \in \mathbb{R}^n$



[Valada et al., 2018]



[Google Translate]

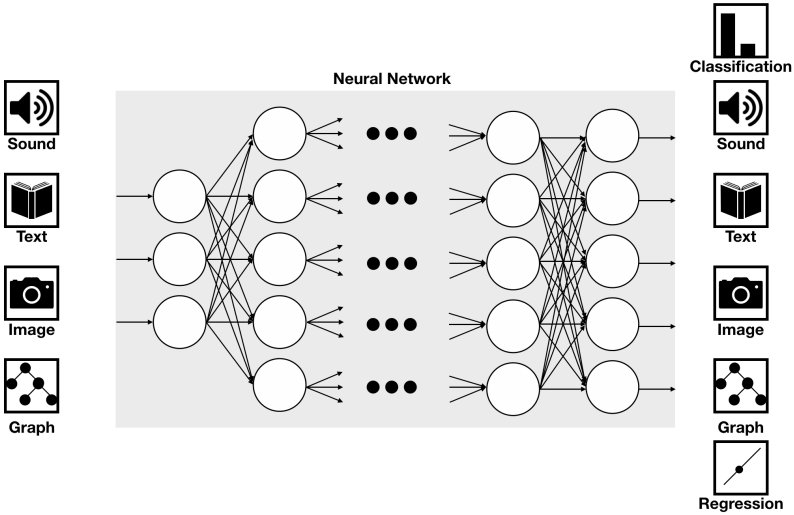


[Chen et al., 2017]



[Chen et al., 2018]

Flexibility of Deep Learning: Structured Inputs & Outputs



Questions to Answer for Yourself / Discuss with Friends

- Repetition: What are some of the input and output modalities that deep learning can handle more easily than standard ML?
- Application to your interests: Which input and output modalities are particularly interesting for your problem of interest?

Lecture Overview

1. Motivation for Deep Learning
2. Representation Learning and Deep Learning
3. Standard ML Concepts We Build On
4. Basics of Neural Networks
5. Flexibility of Deep Learning
6. History of Deep Learning
7. Summary, Further Reading, References

Foundations of Deep Learning

Introduction to Deep Learning

History of Deep Learning

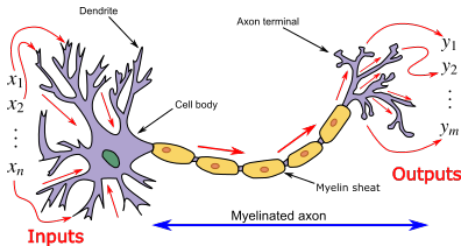
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Biological Inspiration of Artificial Neural Networks

- Dendrites input information to the cell
- Neuron fires (has action potential) if a certain threshold for the voltage is exceeded
- Output of information by axon
- The axon is connected to dentrites of other cells via synapses
- Learning: adaptation of the synapse's efficiency, its synaptical weight



[Image Source: Egm4313.s12 at English Wikipedia via Wikimedia Commons]

History of Deep Learning

Deep Learning has developed in several waves

The early days, under the name of artificial neural networks/cybernetics

- 1943 Artificial neurons as a model of brain function [McCulloch and Pitts, 1943]
- 1949 Hebbian learning [Hebb, 1949]
- 1958 Rosenblatt perceptron [Rosenblatt, 1958]
- 1960 Adaline $\rightarrow$ stochastic gradient descent [Widrow and Hoff, 1960]

The first time the popularity of NNs declined

- Negative result: linear models cannot represent the XOR function
- Backlash against biologically inspired learning [Minsky and Papert, 1969]

History of Deep Learning

1980 - early 2000s (under the name of connectionism)

- 1980 Neocognitron [Fukushima and Miyake, 1980]
- 1986 Multilayer Perceptrons and backpropagation [Rumelhart et al., 1986]
- 1989 Autoencoders [Baldi and Hornik, 1989],
  Convolutional neural networks [LeCun et al., 1989]
- 1997 LSTMs [Hochreiter and Schmidhuber, 1997]

The second time the popularity of NNs declined

- Ventures based on NNs made unrealistically ambitious claims
  - AI research could not fulfill these unreasonable expectations
- Other fields of machine learning made advances
  - E.g., SVMs and graphical models
  - SVMs were the state of the art on many datasets (data was small), specialized ConvNets held state of the art on MNIST but didn't scale

History of Deep Learning and ANNs (cont.)

Mid 2000s, the field got re-invigorated:

- Greedy layer-wise pretraining [Hinton et al., 2006]
  - It was now possible to train much deeper networks
- Several groups "resurrected" the idea of training large neural networks supervisedly using large amounts of data.
  - Most prominently, [Krizhevsky et al., 2012] improved results on Imagenet benchmark by large margin
- Since then: exponential growth
  - NeurIPS attendance has grown exponentially
    - In 2018, it sold out in 12 minutes; they then set up a lottery system
  - Some people are raising unrealistic expectations
  - Let's see how long this current wave persists

Questions to Answer for Yourself / Discuss with Friends

- Repetition: What were the reasons that the popularity of deep learning already declined twice before?
- Discussion: How long will the current deep learning wave persist?
  - What are reasons that it will continue?
  - What are reasons that it will end?

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Summary by Learning Goals

- Deep learning is applied successfully in many fields and is very popular
- Representations of increasing complexity help model the data
- Deep learning reduces the need for preprocessing and can address a variety of tasks
- A single step of training a feedforward neural network consists of a forward pass and a backward pass
- Although the field of deep learning began to develop as early as the 1940s, it was not until mid 2000s that it picked up pace

Further Reading

Read chapter 6 of the Deep Learning Book for a detailed discussion of MLPs.

Check these sites for some of the latest developments in Deep Learning:

- MIT machine learning news
- Deep learning weekly
- AI weekly

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